Predicting Residential Sale Prices in Ames: A Multiple Linear Regression Analysis

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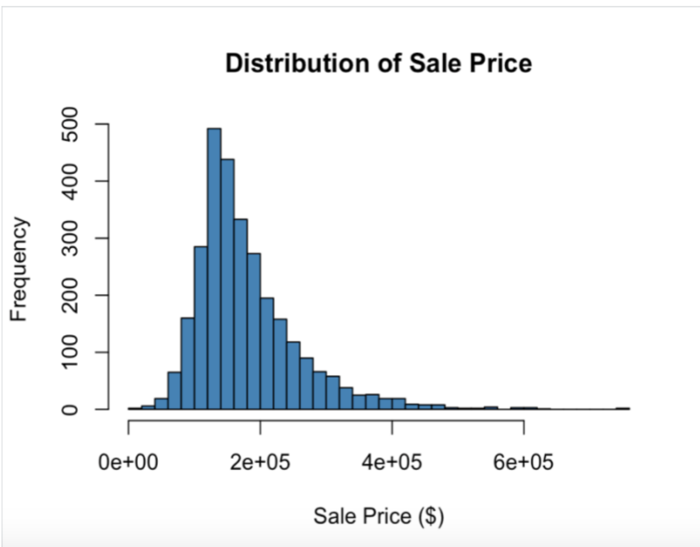
Introduction:

Accurately estimating sales prices for residential properties is a challenge for homeowners and real estate professionals. Traditional appraisal methods rely heavily on the judgment of trained appraisers who compare the property to other recently sold comparable homes. This approach is practical but can be time-consuming, inconsistent, and difficult to scale with the size of the housing market. Multiple linear regression offers a data-driven approach to quantify the statistical relationship between a home's sale price and a set of measurable physical and location-related characteristics. A regression model can generate price predictions, identify which features drive value, and flag properties whose prices deviate unexpectedly from what normal characteristics would predict with the model. The report uses the Ames Iowa housing dataset which contains 2,930 individual residential property sales that occurred between 2006 and 2010 with the 82 recorded variables that describe each property. These variables include lot characteristics, quality ratings, conditions of the sale, and structural attributes, but the outcome variable of interest in this analysis is the SalePrice, which is the final transaction price of each home in US dollars. The purpose of the assignment is to first use exploratory data analysis and descriptive statistics to characterize the data and identify quality issues such as any missing values that must be addressed before modeling. A multiple linear regression model is fit using several continuous predictor variables and the resulting model is interpreted and corrected for problems such as multicollinearity and influential outliers. An all-subsets regression procedure is used to search across a larger pool of candidate predictors for the statistically preferred model. This is then compared to the manually specified model to show which is more appropriate to predict Ames Iowa home prices.

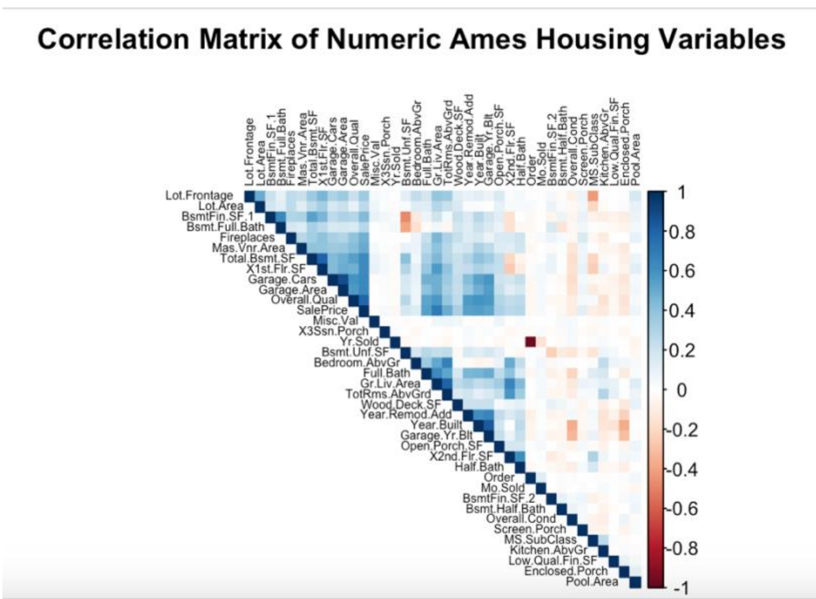
Analysis:

The 82 variables in the Ames housing dataset have 23 nominal, 23 ordinal, 14 discrete, and 20 continuous variables along with two identifier columns (PID and Order). The outcome variable, which was SalePrice, ranged from a minimum of $12,789 to a maximum of $755,000 with the mean being $180,796. The median house price was $160,000 with the standard deviation for the dataset being approximately $79,887. There is a right tail visible in the distribution of sale price which indicates that most homes are sold in a moderate price range, but a smaller number of higher-value properties pull the average upward.



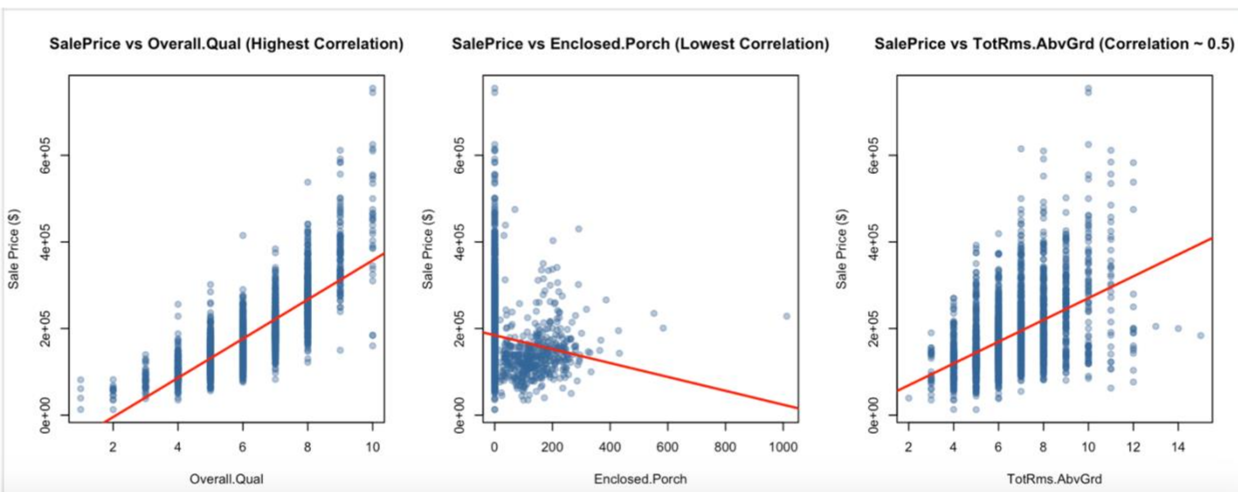
The dataset reveals 13,960 missing cells across 27 of the 82 variables. The large majority of these missing values are not data-entry errors but rather for categorical variables an NA value signifying the physical absence of a feature as opposed to an unrecorded measurement. For example, Pool.QC (pool quality) was missing for a large amount of the properties which corresponds with the number of homes that simply do not have a pool – the same can be said for other variables such as Lot.Frontage, Fence, Alley, Fireplace.Qu. Before modeling, the numeric (continuous and discrete) predictors were replaced with the column mean which is a strategy to avoid discarding observations but may slightly understate the true variability of a variable. For categorical variables, the missing values were recoded as the explicit category “none” which reflects the fact that most of the NAs represent a genuinely absent feature rather than an unknown value.

A correlation matrix was computed for the numeric variables in the dataset using the cor() function. In the plot, each cell represents the Pearson correlation coefficient between a pair of variables. The color intensity indicates the strength of the relationship with dark blue representing strong positive correlation, dark red representing strong negative correlation, and pale or white cells representing correlations near zero.



The variables were reordered using hierarchical clustering so that variables with similar correlation patterns are grouped visually adjacent to one another. This makes clusters of related features (such as the various size measurements or quality ratings) easier to identify at a glance. The Overall.Qual, which is the assessor’s 1-to-10 rating of overall material and finish quality, had the strongest correlation with SalePrice at r = 0.799 which makes it the single most predictive variable in the dataset. On the other end, Enclosed.Porch had the weakest (slightly negative) correlation with the SalePrice at r = -0.129. This suggests that homes with enclosed porches are marginally lower priced on average because enclosed porches are more common in homes that are older and smaller. TotRms.AbvGrd (total rooms above grade excluding bathrooms) had a correlation with SalePrice of r = 0.495 which is the value closest to the target correlation of 0.50 among all numeric predictors

The three scatter plots show how differently these variables all relate to price. The Overall.Qual plot shows a strongly increasing fan-shaped pattern where, as quality rises from 1 to 10, median sale price rises steeply and consistently. However, the spread of prices also widens at higher quality levels since high-quality large homes have much more price variability than modest starter homes.



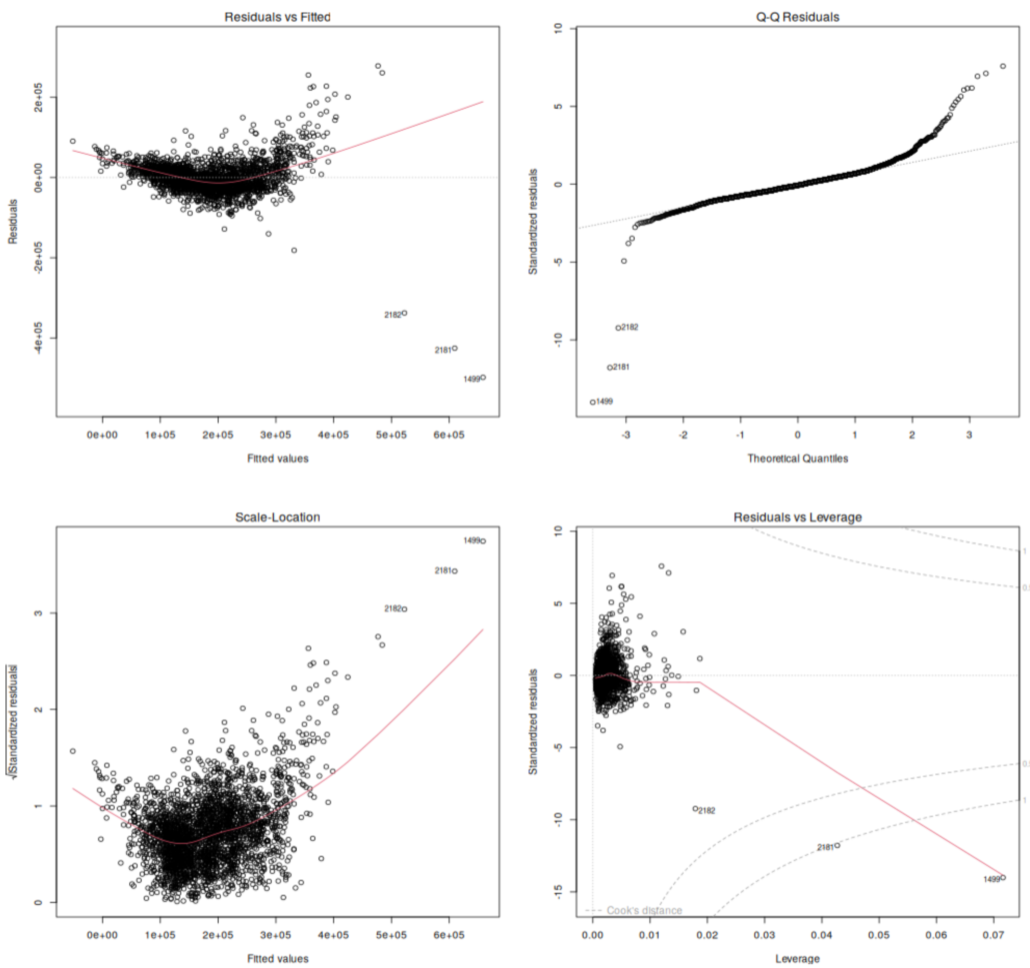
The Enclosed.Porch plot shows no discernible pattern with the majority of homes having an enclosed porch area of zero. The points form a dense vertical cloud near the y-axis with a nearly flat regression line, consistent with the near-zero correlation. The TotRms.AbvGrd plot shows a moderately linear, positive trend where sale price tends to increase as the number of rooms increases, but with a noticeable amount of scatter around the trend line. This reflects that room count is a real but comparatively noisy proxy for overall home size and value.

A multiple linear regression model was fit to predict SalePrice using 5 continuous and discrete numeric predictors chosen for their strong correlations with price. These were above-grade living area (Gr.Liv.Area), total basement square footage (Total.Bsmt.SF), year built (Year.Built), overall quality rating (Overall.Qual), and garage capacity in cars (Garage.Cars). The fitted model (Model 1) can be written in an equation as SalePrice = -699,891.65 + 51.55 (Gr.Liv.Area) + 33.24 (Total.Bsmt.SF) + 313.77 (Year.Built) + 20,809.85 (Overall.Qual) + 13,030.47 (Garage.Cars) + ε

Predictor	Coefficient ($)	Interpretation
Intercept	-699,891.65	Baseline value when all predictors equal zero (not directly interpretable, outside data range)
Gr.Liv.Area	51.55	Each additional square foot of above-grade living area is associated with a $51.55 increase in sale price, holding other variables constant
Total.Bsmt.SF	33.25	Each additional square foot of total basement area is associated with a $33.24 increase in sale price, holding other variables constant
Year.Built	313.77	Each additional year of construction recency is associated with a $313.77 increase in sale price, holding other variables constant
Overall.Qual	20,809.85	Each one-point increase in the overall quality rating (1-10 scale) is associated with a $20,809.85 increase in sale price, holding other variables constant

Garage.Cars	13,030.47	Each additional car of garage capacity is associated with a $13,030.47 increase in sale price, holding other variables constant

The base R plot() function was applied to Model 1 to produce 4 standard diagnostic plots which are shown in the figure below: Residuals vs. Fitted, Normal Q-Q, Scale-Location, and Residuals vs. Leverage

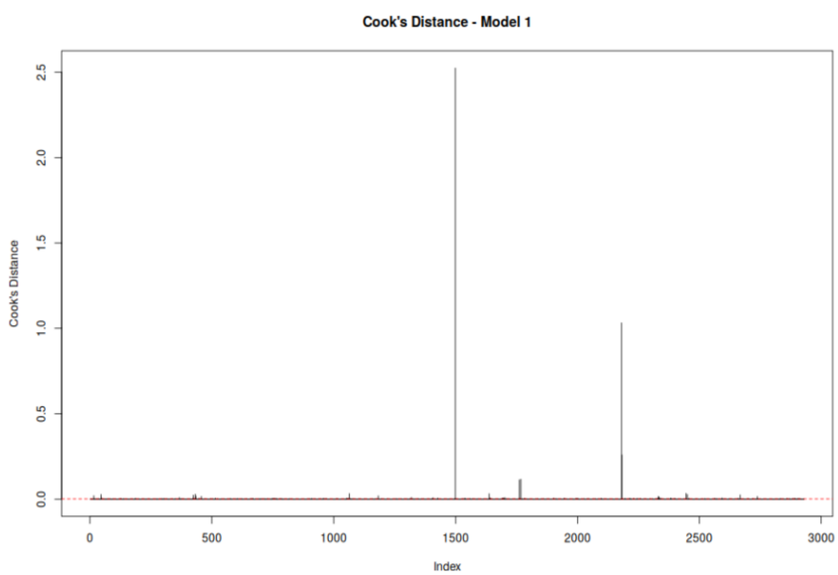


The residual vs. fitted plot shows a mild funnel shape with residual spread increasing at higher fitted sales prices, which is evidence of heteroscedasticity, a violation of the constant-variance assumption of ordinary least squares regression. The Normal Q-Q plot follows the diagonal reference line through the middle of the distribution but deviates noticeably in the upper right tail. This indicates that the highest-price homes have larger right-skewed residuals than a normal distribution would predict. The Scale-Location plot reinforces the heteroscedasticity finding which shows an upward sloping trend in the square root of standardized residuals as fitted values increase. The Residuals vs. Leverage plot identifies several points with unusually high leverage or large Cook's distance. This is noticeable for a small number of very expensive homes that sit far from the majority of this data. These plots suggest Model 1 suffers from non-constant error variance and a small number of influential high-price observations.

Multicollinearity, the presence of strong linear relationships among predictor variables, can inflate standard errors of regression coefficients which make them difficult to interpret. Multicollinearity was assessed for Model 1 using the variance inflation factor (VIF) with the general rule being VIF values above 5 indicate moderate concern and values above 10 indicating serious multicollinearity.

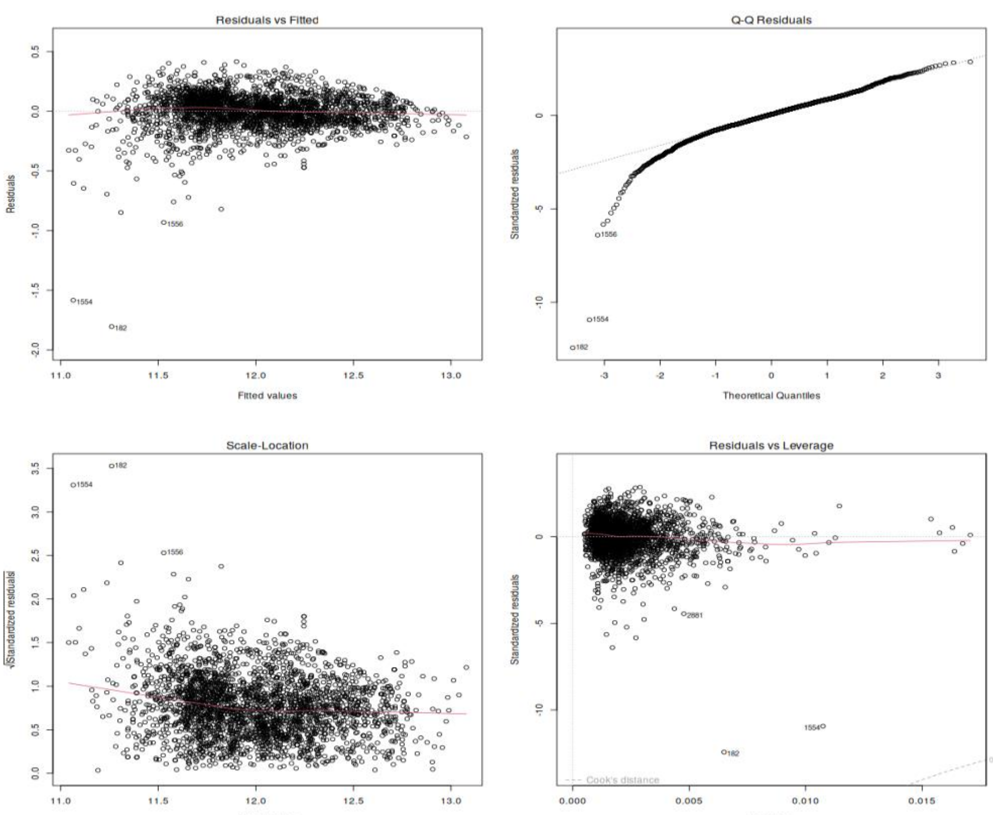
Predictor	VIF
Gr.Liv.Area	1.70
Total.Bsmt.SF	1.53
Year.Built	1.80
Overall.Qual	2.48
Garage.Cars	1.85

All of the VIF values fell below 2.5 and this indicates that multicollinearity is not a meaningful problem for Model 1 since the five predictors are not strongly correlated with one another. This result makes sense given that living area, basement size, age, quality rating, and garage capacity capture largely distinct aspects of a home. In order to detect outliers and influence diagnostics, standardized residuals were examined and any observation with a residual greater than 3 was flagged as a statistical outlier which included 40 observations. Cook's distance, which measures how much the fitted regression coefficients would change if a given observation were removed, was computed for every observation – this identified 168 observations as unusually influential



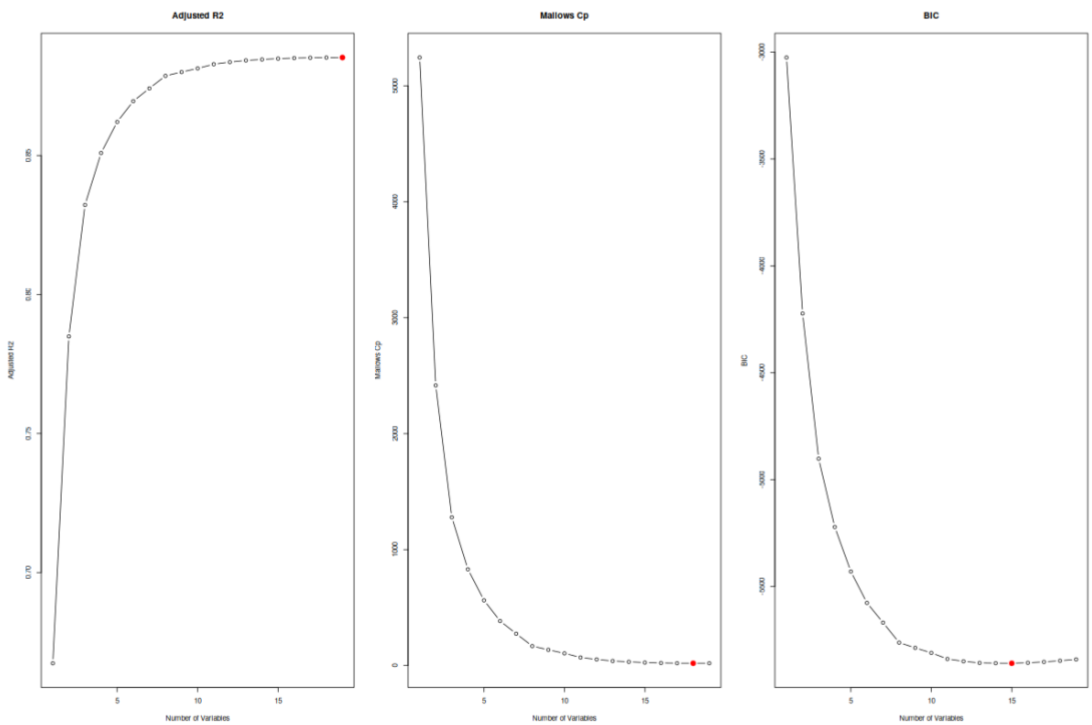
High-leverage points correspond primarily to very large or expensive homes which includes a number of properties with very large living areas related to their sale price. This reflects a small number of atypical sales such as homes sold below market value to family members or large homes in lower-priced neighborhoods. These observations are not data-entry errors, but they exert disproportionate influence on the fitted regression line and contribute to the heteroscedasticity observed in the plots. 168 observations represent under 6% of the sample and their present distorts model fit, so removing the most influential subset would be an appropriate correction.

To correct the problems identified above, the 168 observations flagged by Cook's distance were removed from the dataset which reduced the sample from 2,930 to 2,762 observations. Since the diagnostic plots showed evidence of heteroscedasticity and right-skewed residuals tied to the right skew of SalePrice itself, the outcome variable was log-transformed. The resulting corrected model, Model 2, uses the same five predictors as Model 1. SalePrice = 6.298 + 0.000293 (Gr.Liv.Area) + 0.000198 (Total.Bsmt.SF) + 0.00222 (Year.Built) + 0.0982 (Overall.Qual) + 0.0589 (Garage.Cars) + ε.



Model 2 showed a noticeable improvement in fit since R^2 rose from 0.787 to 0.847 meaning the corrected model now explains roughly 84.7% of the variance in log sale price. The Residuals vs.

Fitted plot is much flatter, the Normal Q-Q plot tracks the reference line more closely and the Scale-Location plot shows a much flatter trend which indicates reduced heteroscedasticity. To determine whether a better performing model exists among the pool of predictors, an all-subsets regression was performed using the regsubsets function from the leaps package.



This figure adjusts R^2, Mallows' Cp, and Bayesian Information Criterion (BIC) across all candidate model sizes. The criteria disagreed slightly on the ideal number of predictors with R^2 improving up to the full 19 variable model, Mallows Cp was minimized at 18 variables, and BIC was minimized at the 15-variable model. The BIC model was selected as the "best" all-subsets model for comparison purposes because it tends to protect against overfitting and produce more generalizable models. The model equation could be written as SalePrice = -1,147,000 + 0.626 (Lot.Area) + 14,380 (Overall.Qual) + 4,463 (Overall.Cond) + 354.0 (Year.Built) + 194.6 (Year.Remod.Add) + 16.47 (Mas.Vnr.Area) + 29.60 (Total.Bsmt.SF) + 60.75 (X1st.Flr.SF) +

57.46 (X2nd.Flr.SF) – 3,516 (Full.Bath) – 5,325 (Bedroom.AbvGr) + 4,169 (Fireplaces) + 34.77 (Garage.Area) + 17.12 (Wood.Deck.SF) + 19.91 (Open.Porch.SF) + ε. The model achieved an adjusted R^2 of 0.885 with all 15 predictors statistically significant at $p < 0.01$.

Model 2 (corrected, 5 predictor log model) and the Best-Subset Model (15 predictor BIC model) differ substantially in complexity and accuracy. Since the two models use different outcome scales (log dollars vs dollars), their AIC and BIC values are not directly comparable. Model 2's log-scale predictions can be exponentiated back to the dollar scale and both models' predictions can be compared to sale prices using RMSE and MAE.

Model	Predictors	Adjusted R^2	RMSE ($)	MAE ($)
Model 2 (log, outliers removed)	5	0.847	22,262	16,932
Best-Subset Model (BIC)	15	0.885	21,945	16,800

On a common dollar scale, the Best-Subset Model's RMSE ($21,945) is only about $317 lower than Model 2's back-transformed RMSE ($22,262) and their MAE values differ by $132. The Best-Subset Model does have real advantages because it captures additional, interpretable value drivers that Model 2 entirely omits such as lot size, remodel recency, fireplace count, and outdoor living space. These are factors that real estate professionals expect to affect housing prices. The BIC Model is preferred for a use case where explanatory completeness and accuracy gains matter such as assessing value for a specific property. However, if the property is simpler and more easily communicated, the five-predictor structure remains an attractive alternative.

Conclusion:

Descriptive statistics and correlation analysis of the Ames Housing dataset confirmed that overall quality rating, above-grade living area, and total basement square footage are the strongest individual predictors of sale price. The five-variance regression model explains roughly 79% of the variance in sale price but showed evidence of heteroscedasticity and a meaningful cluster of influential high-price outliers. Correcting these issues by removing these influential observations and log-transforming the skewed sale-price outcome produced a model with an adjusted R^2 value of 85%. The all-subsets regression search across the pool of 19 predictors decided on the 15-variable BIC model that achieved the overall strongest fit. More broadly, this report illustrates why model building is an iterative process as opposed to a single step with corrections and expanding predictor variables measurably improving statistical validity and practical usefulness of the final model's results. In the future, this analysis could be extended by incorporating categorical predictors such as neighborhood or exploring interaction terms such as the relationship between overall quality and living area.

References

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